

Lecture 7 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 7 introduces normal models for continuous biomedical data. The lecture moves from the one-parameter binary models used earlier in the course to likelihoods with both a center and a spread. It then applies the Bayesian workflow to a normal model for normalized whole brain volume, extends the model to compare age groups, and introduces raw and normalized effect sizes for normal models. The lecture ends by distinguishing unpaired and paired comparisons, showing how outliers motivate robust Student's t likelihoods, and connecting Bayesian paired analyses to classical paired tests.

Lecture Sections

7.1 — Continuous data

This section explains why continuous measurements require different likelihoods from the binary and count data used earlier in the course. Continuous biomedical variables come with units, have both a middle and a spread, and may be bounded or multimodal. The normal likelihood is introduced as the basic model for continuous data, with one parameter for the population center and one for the population variability.

New Definitions and Terms

- **Continuous data** — measurements represented on a numerical scale with units, such as reaction time, blood pressure, or normalized brain volume
- **Normal model** — a model for continuous data using a normal likelihood, usually parameterized by a mean and a standard deviation
- **Mean parameter** — the parameter representing the center of the modeled population
- **Scale parameter** — the parameter representing the spread of the modeled population

Formulas

$$y_i \sim \text{Normal}(\mu, \sigma)$$

Each observed continuous value is modeled as a draw from a normal distribution with population mean μ and population standard deviation σ .

$$\mu \sim \text{Normal}(\mu_\mu, \sigma_\mu), \quad \sigma \sim \text{HalfNormal}(\sigma_\sigma)$$

The mean can have a normal prior, while the standard deviation needs a positive prior such as a half-normal.

Figure

Likelihoods of biomedical data

- All data in lectures so far has been binary
 - Tutorial has done counts as well
- Likelihoods have had only one parameter
- Changing the type of data requires changing the likelihood

Binary	Grouped	Ordered Groups	Counts	Continuous
Survived / died	Blood type	Pain score	Number of falls	Reaction time (ms)
Patient / control	Cell type	Sleep quality	Spike count	Voice pitch (Hz)
Mutation present	Diagnosis	Disease stage	# of seizures	Cerebellar volume
Drug responder	Sleep stage	Tumor grade	# of mutations	Blood pressure
Seizure occurred	Strategy type	Confidence rating	# of infections	Movement error
Binomial	Multinomial	Ordered logistic model	Poisson	Normal

Figure 1: Slide 4 — Likelihoods of biomedical data

Translate the model to PyMC

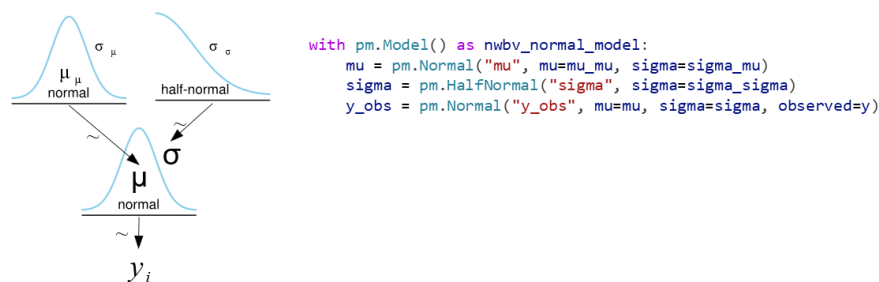


Figure 2: Slide 8 — Translate the model to PyMC

7.2 — Normal model workflow

This section applies the Bayesian workflow to the normal model including prior predictive check before seeing the data. After sampling, trace plots and diagnostics are used to check sampling quality, and posterior predictive checks are used to evaluate model fit. Bayesian p values are introduced as a way to quantify how well posterior predictive data resemble the observed data.

New Definitions and Terms

- **Cumulative distribution** — a representation showing the probability of values less than or equal to each value on the x-axis
- **Bayesian p value** — the posterior predictive probability that a simulated statistic is as extreme as, or more extreme than, the observed statistic
- **Posterior predictive sampling distribution** — the distribution of simulated summaries generated from posterior predictive datasets

Formulas

$$p_B = P(T(y^{rep}) \geq T(y) \mid y)$$

A Bayesian p value compares a statistic calculated on posterior predictive datasets, $T(y^{rep})$, with the same statistic calculated on the observed data, $T(y)$.

Figure

Posterior predictive data matches real data

- Look at both posterior density and cumulative distribution

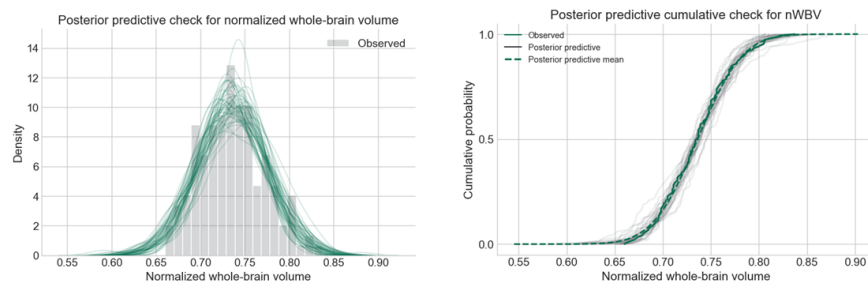


Figure 3: Slide 16 — Posterior predictive data matches real data

Arviz will show Bayesian p values of different sample statistics

- The question is always:
 - How well does the fit model produce data like my data

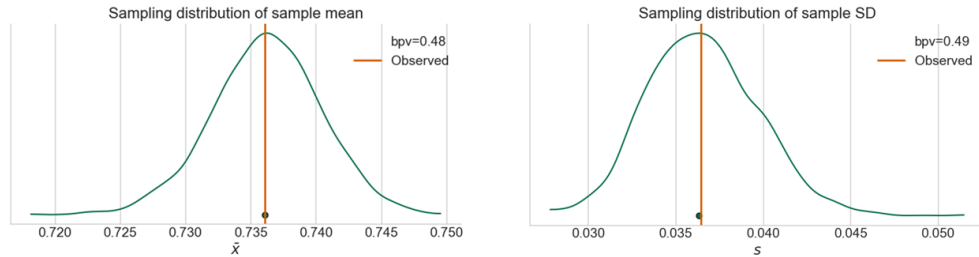


Figure 4: Slide 20 — Arviz will show Bayesian p values of different sample statistics

7.3 — Comparing across ages

This section extends the normal model of brain volume to compare brain volumes of two age groups. The model is a pooled / unpooled: each group has its own mean, while they use a shared standard deviation. The main quantity of interest is the difference between the means of the two groups.

New Definitions and Terms

- **Group index** — an integer or categorical variable used to select the correct group-specific parameter for each observation
- **Shared standard deviation** — an assumption that both groups have the same population spread
- **pm.Data** — a PyMC data object that is stored in the model and can be updated later

Formulas

$$y_i \sim \text{Normal}(\mu_{g_i}, \sigma)$$

Each observation uses the mean corresponding to its group g_i , while sharing the same standard deviation σ .

$$\Delta_\mu = \mu_1 - \mu_0$$

The difference between group means is stored as a deterministic parameter so it can be examined directly in the posterior.

Figure

Translate model to PyMC

- Unpooled model

```
with pm.Model(coords=coords) as group_model:
    group = pm.Data("group", group_idx, dims="obs")
    mu_group = pm.Normal("mu_group", mu=mu_mu, sigma=sigma_mu, dims="group")
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu_diff = pm.Deterministic("mu_diff", mu_group[1] - mu_group[0])
    nwbv_obs = pm.Normal("nwbv_obs", mu=mu_group[age_group], sigma=sigma,
                        observed=nwbv_age, dims="obs")
```

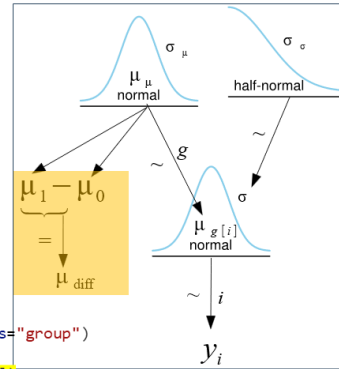


Figure 5: Slide 26 — Translate model to PyMC

7.4 — Effect sizes for normal models

This section distinguishes raw and normalized effect sizes for a normal two-group comparison. The raw effect size is the difference in measured units, here percent total intracranial volume. The normalized effect size is Cohen's d , which divides the mean difference by a standard deviation and therefore depends on the variance structure assumed by the model.

New Definitions and Terms

- **Cohen's d** — a normalized mean difference, here calculated as a posterior deterministic quantity
- **Student's t distribution** — a distribution with heavier tails than the normal distribution, controlled by a degrees-of-freedom parameter
- **t test** — a frequentist test comparing a mean difference or standardized mean difference to a t sampling distribution

Formulas

$$\Delta_{\mu} = \mu_1 - \mu_0$$

The raw effect size is the difference between the two group means, measured in the original units.

$$d = \frac{\Delta_{\mu}}{\sigma}$$

Cohen's d is the mean difference divided by the standard deviation. In this lecture's model, the denominator is the single shared σ .

Figure

Effect sizes for the difference between two groups

- Raw effect size
 - The difference in measured units
 - Percent tICV
- Normalized effect size
 - Cohen's d
 - Assumes a single variance in the denominator
 - If we had modeled two variances
 - The formula would be different

$$\text{Cohen's } d = \frac{\mu_1 - \mu_2}{\sigma}$$

$$\text{or } \frac{\mu_1 - \mu_2}{\sqrt{\frac{\sigma_1^2 + \sigma_2^2}{2}}}$$

Effect size	Cohen's d
None	0
Small	0.2
Medium	0.5
Large	0.8

Figure 6: Slide 33 — Effect sizes for the difference between two groups

Both raw and normalized effect sizes have distributions

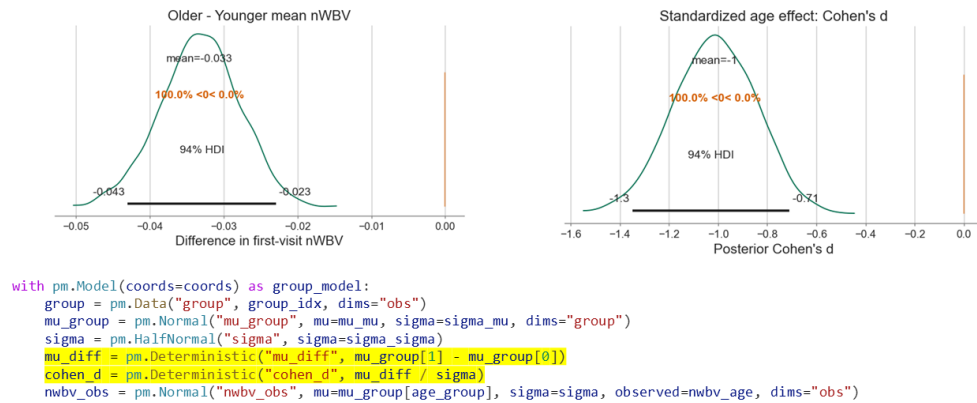
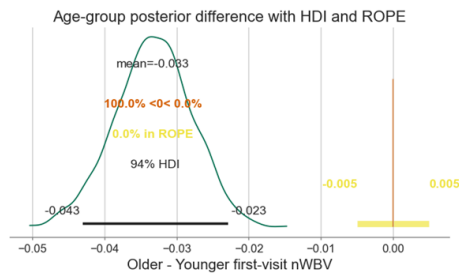


Figure 7: Slide 34 — Both raw and normalized effect sizes have distributions

A frequentist test for comparing two groups

- The t test
 - Compares frequentist Cohen's d
 - To the t distribution
 - With the correct degrees of freedom

Bayesian hypothesis test for comparison



Statistic	Value
Younger mean	0.753686
Older mean	0.720662
Younger SD	0.035784
Older SD	0.029551
Pooled SD	0.032605
Cohen's d (Older - Younger)	-1.012819
Degrees of freedom	148.000000
94% two-sided t threshold	1.895319
Two-sided p-value	0.000000

Figure 8: Slide 39 — A frequentist test for comparing two groups

7.5 — Paired comparisons

This section contrasts unpaired comparisons between groups with paired comparisons within the same individuals. In the MRI example, each participant has several scans, so the relevant question can be asked as the change in normalized brain volume from the first to the last visit. Pairing removes much of the between-subject variability and can reveal a consistent change that is harder to see in an unpaired comparison.

New Definitions and Terms

- **Paired comparison** — a comparison made within the same subjects or units across two conditions or times
- **Unpaired comparison** — a comparison made between separate groups of observations
- **Within-subject change** — the difference between two measurements from the same subject
- **Between-subject variability** — variability among people that can hide consistent within-person effects
- **Outlier** — an observation that is poorly explained by the fitted model and can strongly influence a normal model

Formulas

$$\text{change}_i = y_{i,\text{last}} - y_{i,\text{first}}$$

The paired analysis converts repeated measurements into one change value per subject.

$$\text{change}_i \sim \text{Normal}(\mu_{\text{change}}, \sigma_{\text{change}})$$

The paired normal model is just the normal model applied to the vector of within-subject changes.

Figure

Variability between subjects can hide consistency

- Almost all subjects
 - Had smaller normalized brain volume
 - On the last visit

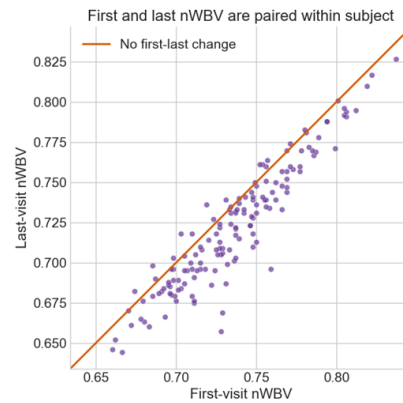
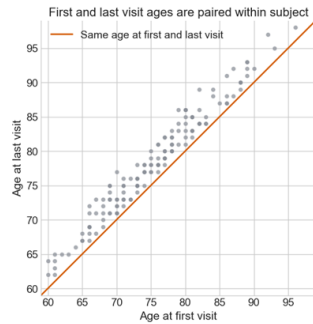


Figure 9: Slide 43 — Variability between subjects can hide consistency

7.6 — Robust models

This section uses the failure of the paired normal model to motivate a robust likelihood. Normal likelihoods are sensitive to outliers because they assign very low probability to observations far from the mean. Replacing the normal likelihood with a Student's t likelihood adds a degrees-of-freedom parameter, allowing the model to have thicker tails when the data require them.

New Definitions and Terms

- **Robust model** — a model whose conclusions are less sensitive to outliers or deviations from ideal assumptions
- **Student's t likelihood** — a continuous likelihood with heavier tails than a normal likelihood
- **Degrees of freedom parameter** — the parameter ν controlling the tails of the Student's t distribution
- **Normality parameter** — the lecture's interpretation of ν : small values imply heavier tails, while large values converge toward a normal model
- **Thick tails** — a distributional shape that assigns more probability to extreme values

Formulas

$$\text{change}_i \sim t_\nu(\mu_{\text{change}}, \sigma_{\text{change}})$$

The robust paired model replaces the normal likelihood with a Student's t likelihood.

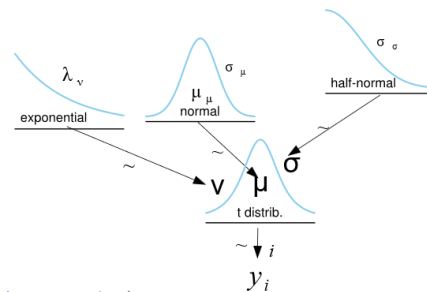
$$\nu = \nu_{\min} + \nu_{\text{excess}}, \quad \nu_{\text{excess}} \sim \text{Exponential}(\lambda)$$

Adding a lower bound keeps the tails from becoming unrealistically wide and helps avoid sampling problems.

Figure

As Bayesians, we can choose our model

- Replace normal likelihood with Student's t likelihood
- One more parameter
 - The 'normality' parameter
 - If posterior is small : robust to outliers
 - If posterior is large : converge to normal model



```
with pm.Model() as paired_student_t_model:
    mu_change = pm.Normal("mu_change", mu=mu_mu_student, sigma=sigma_mu_student)
    sigma_change = pm.HalfNormal("sigma_change", sigma=sigma_sigma_student)
    nu = pm.Exponential("nu", lam=lam_nu_student)
    change_obs = pm.StudentT("change_obs", nu=nu, mu=mu_change, sigma=sigma_change, observed=change_paired)
```

Figure 10: Slide 50 — As Bayesians, we can choose our model

7.7 — Effect sizes for paired comparisons

This section returns to effect sizes after fitting the robust paired model. The raw paired effect is the mean change, and the normalized effect is the paired Cohen's d . The lecture also shows that deterministic quantities can be added to the posterior after sampling by operating directly on XArray objects in the `idata` object.

New Definitions and Terms

- **Paired raw effect** — the posterior distribution of the mean within-subject change
- **Paired Cohen's d** — the mean within-subject change divided by the standard deviation of the within-subject changes
- **XArray DataArray** — the object type used inside ArviZ inference data for posterior variables
- **Broadcasting** — automatic alignment of array operations across compatible dimensions
- **Wilcoxon signed-rank test** — a frequentist paired test that can be used when a normal likelihood is not appropriate

Formulas

$$d_{paired} = \frac{\mu_{change}}{\sigma_{change}}$$

The normalized paired effect size is calculated from posterior variables.

```
idata.posterior["cohen_d"] = (  
    idata.posterior["mu_change"] / idata.posterior["sigma_change"]  
)
```

A deterministic posterior quantity can be added after sampling if it is just a calculation on sampled posterior variables.

Figure

Compare paired and unpaired analyses

- We can add a calculation to the posterior if we didn't sample it

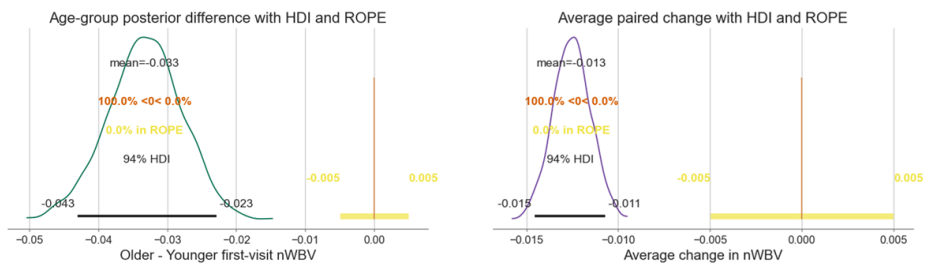


Figure 11: Slide 59 — Compare paired and unpaired analyses

Tests on paired data

- The t-test assumes that the likelihood is normal
 - Not appropriate for non-normal data
- The Wilcoxon signed-rank test can be used

Quantity	Value
Sample mean of paired difference	-0.013400
Sample SD of paired difference	0.013600
Cohen's dz	-0.987500
Degrees of freedom	149.000000
94% two-sided t threshold	1.895200
Two-sided p-value	0.000000

Quantity	Value
W statistic	603.500000
94% two-sided W lower threshold	4402.770200
Two-sided p-value	0.000000

Figure 12: Slide 61 — Tests on paired data

7.8 — Next week: linear models

This final section connects normal models to the next topic: linear models. Up to this point, the course has mostly modeled one outcome variable at a time, sometimes with grouping variables. Linear models extend this framework to relationships among variables, allowing questions about how one measurement changes with another.

Figure

Today and next week

- Up until now
 - Always only one variable
 - Except grouping variables
- From next week
 - How different variables are related to each other
 - Height and weight
 - Intelligence and brain size
 - Political affiliation and musical taste
 - Gene expression and risk of Parkinson's
 - Neural activity and intended direction of movement

Figure 13: Slide 65 — Today and next week